

Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

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Abstract—Continuous wearable EEG is increasingly paired with consumer wearables that provide a heartbeat stream (e.g., smartwatch ECG/PPG). Taking the available ECG/heartbeat reference, we propose a repetitive-control framework that models cardiac artifact as a heartbeat-aligned periodic disturbance. Each EEG sample is assigned a cardiac phase index wrapped around ECG R-peaks; phase-synchronous averaging then preserves heartbeat-correlated residue while attenuating non-phase-locked noise. Subtracting this phase-indexed average yields effective artifact removal in the non-causal offline setting. For online real-time operation, an iterative-learning estimator is used to track the residue in either a same-beat predictive mode (strict causal) or a one-beat-delayed mode (buffered causal). We also propose a PI based phase tracker for robust heartbeat timing estimation from raw EKG waveforms to account for breathe related heart rate variability or immature beats rejection. Results on open datasets and semi-synthetic simulations show that the heartbeat-timing-anchored repetitive/iterative learning approach provides strong artifact suppression at low compute cost, making it well suited to always-on wearable execution, where blind source separation is typically channel-starved.

I. INTRODUCTION

Wearable EEG is moving toward continuous, day-scale monitoring, and many users already wear a second device that provides heartbeat timing from ECG or PPG. This creates a practical path for low-channel ear-EEG denoising: use heartbeat timing already available from the wearable ecosystem rather than requiring additional EEG channels. In ear-centered montages, cardiac field artifact (CFA) is often prominent because electrodes are close to neck and vascular pathways. CFA is the volume-conducted electrical field of the heart [1], distinct from the scanner-related ballistocardiogram artifact emphasized in EEG-fMRI work [2]. For out-of-scanner earables, CFA is therefore a primary contamination mechanism.

Classical ECG-referenced adaptive cancellation is well established [3], and it is a natural baseline. However, in our target regime it faces two practical limits: earables are channel-starved for blind source separation [6], and reference-to-artifact waveform predictability can be weak under real volume-conduction mismatch and nonstationarity. Our real-data observations in this paper (Fig. 1) support exactly this point: coherence between ECG and ear-EEG artifact is close to

zero even with beat-synchronous pooling, so reference-driven waveform regression is limited as a primary mechanism. At the same time, beat timing remains highly informative, because the artifact is still phase-locked to R-peaks.

This motivates a timing-only control view: model cardiac contamination as a repetitive disturbance and reject it by heartbeat phase indexing. Repetitive control and iterative learning control (ILC) are mature tools for periodic disturbance rejection in mechatronics [8], and they map naturally to the heartbeat cycle when robust R-peak timing is available. We therefore use a min/max envelope tracker with PI/PID phase tracking and arrhythmia-aware supervision to produce stable beat timing, then apply phase-aligned template learning/cancellation in three deployment modes: non-causal offline, buffered causal (one-beat latency), and strict real-time causal (zero added latency).

The resulting contribution is a compact, wearable-oriented CFA removal scheme aligned with realistic device constraints: low channel count, low compute budget, and timing references that may come from a separate wearable. We evaluate the design on open real recordings and semi-synthetic data, compare against linear RLS-based and ICA baselines, and quantify the latency-versus-suppression tradeoff between non-causal, buffered, and strict-causal operation.

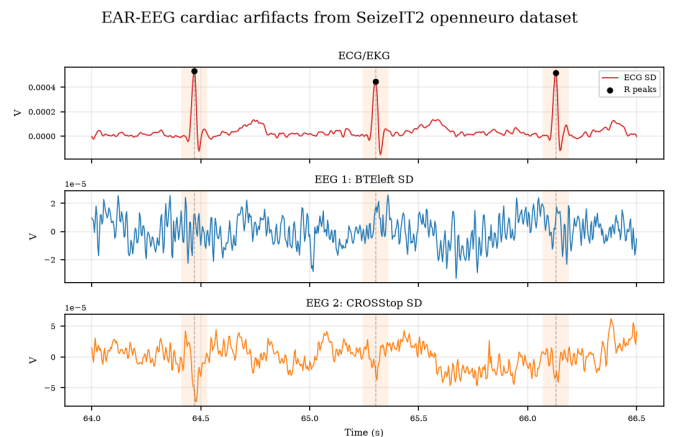


Fig. 1. Short aligned segment from SeizeIT2 (ds005873, sub-001, ses-01, run-01), showing chest ECG/EKG and two ear-EEG channels over about three heartbeats. R-peak neighborhoods are highlighted to emphasize heartbeat-locked contamination timing. For this panel, MNE preprocessing is applied to EEG channels only: 50 Hz notch plus 0.5 Hz high-pass (DC-removal) filtering.

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II. PROBLEM FORMULATION

Following repetitive-control modeling conventions [8], [9], we model the measured ear-EEG signal as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. In ear-EEG, CFA is dominated by the repetitive component $d_r(t)$. R-peak phase-asynchronous disturbances and other channel noise are represented by $n(t)$, and $n(t)$ is assumed zero-mean white noise. DC offset and slow drift are removed by preprocessing.

Assumption 2. The beat period T_k is slowly varying, dominated by respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.

Assumption 3. In the repetitive disturbance rejection and iterative learning estimator below, R-peaks are assumed to be accurately detected and phase-locked. A robust min/max-envelope PI tracker used for that locking is provided in the next section.

Using heartbeat phase, we represent the repetitive disturbance as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the R-peak-aligned cardiac phase. Equivalently, within each beat interval the disturbance is periodic in local time with period T_k , while T_k itself evolves slowly across beats. Here t denotes the discrete-time index governed by the EEG sampling rate, and we assume the ECG reference is sampled at the same rate as EEG.

Problem 1. Finding the optimal estimate of the repetitive template $\hat{d}_r$. We assign each phase-aligned EEG sample a periodic phase index ϕ_p over an R-peak-centered template window,

$$\hat{d}_r(\phi_p) = \frac{1}{N_r} \sum_{k=1}^{N_r} y_k(\phi_p), \quad (4a)$$

$$p = -n_{\text{pre}}, \dots, -1, 0, 1, \dots, n_{\text{post}}, \quad (4b)$$

$$n_t = n_{\text{pre}} + n_{\text{post}}, \quad (4c)$$

where n_t denotes the template-window length tied to the sampling rate, N_r is the number of R-peaks over the recording session, and $y_k(\phi_p)$ is the phase-aligned EEG sample at phase index ϕ_p from beat k ; $p = 0$ is aligned with the R-peak. Stacking $\hat{d}_r(\phi_p)$ over all p yields a continuous template carried

across all beats in the recording. The corresponding subtraction for all same-phase-synchronized EEG samples is

$$\hat{s}(t) = y(t) - \hat{d}_r(\hat{\phi}(t)). \quad (5)$$

Note 1. Eq. (4a) is the non-causal phase-synchronized averaging estimate over the full recording, and the resulting template is used for subtraction in Eq. (5).

Note 2. Based on Assumption 1, in the Eq. (4a) estimate, $n(t)$ is rendered toward zero as the sample mean of zero-mean white noise.

Note 3. A practical guideline for choosing n_{pre} , n_{post} , and n_t : set $n_{\text{pre}} = \lfloor 0.05 f_s \rfloor$ (about 50 ms before the R-peak, covering QRS onset) and $n_{\text{post}} = \lfloor 0.55 f_s \rfloor$ (about 550 ms after, covering the full QRS-T complex at resting heart rate), giving $n_t = n_{\text{pre}} + n_{\text{post}} \approx 0.6 f_s$ samples, i.e. 600 ms total window length. At $f_s = 250$ Hz this yields $n_t = 150$ samples.

Problem 2. Estimating beat timing robustly, i.e., the R-peak index k and R-peak instant, so that EEG samples can be phase-aligned. We denote estimated R-peaks by $\hat{t}_k$, estimated beat period by $\hat{T}_k = \hat{t}_{k+1} - \hat{t}_k$, and estimated phase by

$$\hat{\phi}(t) = \frac{t - \hat{t}_k}{\hat{T}_k}, \quad t \in [\hat{t}_k, \hat{t}_{k+1}). \quad (6)$$

The next section addresses this timing-estimation problem with a min/max-envelope PI tracker.

Note 4. Since T_k is slowly varying (Assumption 2), $\hat{T}_k$ can be reliably predicted between successive R-peaks. The PI tracker operates as a phase-locked loop (PLL): it uses detected R-peak fiducials to correct the instantaneous rate estimate and advances phase continuously between beats, so that $\hat{\phi}(t)$ remains well-defined even in real-time causal operation where the next R-peak has not yet been observed.

III. REPETITIVE DISTURBANCE REJECTION

Given the phase-indexed disturbance model in Sec. II, we perform cancellation by repetitive disturbance rejection with iterative learning on heartbeat-aligned samples. The cancellation target is the beat-synchronous component of CFA, while asynchronous components are left in the residual.

A. ILC/repetitive template cancellation

A template $m(o)$, indexed by time-since-R offset o , stores d_p . Cancellation is a table lookup at the current offset,

$$e_I(t) = y(t) - m(o(t)), \quad (7)$$

and learning is a first-order ILC (leaky-integrator) update,

$$m(o) \leftarrow m(o) + a e_I(t), \quad (8)$$

applied only on normal beats, with a the ILC learning gain. Because we estimate an additive disturbance rather than invert a plant, the effective plant is the identity, and the ILC convergence condition reduces to $|q(1 - a)| < 1$ for a Q -filter/forgetting factor $q \in (0, 1]$ —trivially satisfied for any sane a . The factor q trades asymptotic cancellation for drift tracking.

Detection latency forces a mode choice: the R-peak is confirmed ~ 60 – 80 ms after it occurs, so *buffered* operation (≈ 1 -beat latency) anchors each sample to its detected t_k and removes the sharp R spike fully, whereas *causal* (zero-latency) operation must anchor to the PLL-predicted R-time; the R-spike depth is then capped by RR predictability (RSA-lag dominated). The broad P and T waves survive prediction; only the R spike is latency-critical.

B. Safeguards against cancellation of neural signal

Because $s(t)$ can share frequency content with $d(t)$, frequency alone cannot separate them; the synthesis instead relies on *phase-lock* and *reference-correlation*, together with two explicit limiters.

Phase-lock gate (structural). The ILC template (8) only accumulates energy that is consistent across many beats at the same offset o —it is structurally an evoked average. EEG activity that is not phase-locked to the R-peak averages toward zero in $m(o)$ rather than being learned, regardless of its frequency. Cancellation (7) is further confined to samples inside the template window $o(t) \in [0, L_w]$; outside it, $y(t)$ is untouched.

Amplitude limiter. To bound worst-case over-subtraction, cancellation is capped using the already-computed peak tracker $pk(t)$ from (10):

$$e_I(t) = y(t) - \text{sat}_{\pm\beta pk(t)}(m(o(t))), \quad \beta \in (0, 1), \quad (9)$$

where $\text{sat}_{\pm L}(x) = \max(-L, \min(L, x))$. The subtracted amount can never exceed a fraction β of the tracked ECG amplitude, a physically motivated ceiling on plausible CFA magnitude.

Remaining risk. The heartbeat-evoked potential (HEP) is genuine neural activity that is phase-locked to the heartbeat [11]; phase-lock alone cannot distinguish it from CFA, and it is only partially protected by (9). We report alpha-rhythm preservation (Sec. VII) as an indirect check; an explicit HEP-preservation test is future work.

IV. ECG PHASE ESTIMATOR

A. Min/max tracker + PI/PID phase tracking

A QRS-emphasis band-pass feeds two streaming trackers: a peak tracker $pk(t)$ (fast attack, slow release) and a baseline tracker $bl(t)$ (slow moving average). The normalized gate

$$g(t) = \frac{x(t) - bl(t)}{pk(t) - bl(t)} \quad (10)$$

is amplitude-drift invariant, so posture/contact drift does not defeat detection. A Schmitt hysteresis and a 250 ms refractory yield clean beat gating; the fiducial t_k is the raw-reference maximum within the gated window (low jitter).

A PI phase-locked loop converts sparse fiducials into a continuous phase. Between beats the phase advances by the instantaneous rate estimate,

$$p(t+1) = [p(t) + \text{fr}(t)] \bmod 1, \quad \text{fr}(t) = \frac{1}{\text{RR}_{\text{est}}(t) f_s}, \quad (11)$$

where f_s is the sample rate. At each fiducial, a proportional term realigns phase and an integral term updates the rate estimate,

$$p \leftarrow p - K_p e_p, \quad (12)$$

$$\text{RR}_{\text{est}} \leftarrow \text{RR}_{\text{est}} + K_i (\text{RR}_{\text{meas}} - \text{RR}_{\text{est}}), \quad (13)$$

with e_p the wrapped phase error and K_p, K_i the PI gains. The loop bandwidth is set above respiratory sinus arrhythmia (RSA, ~ 0.2 – 0.4 Hz) and below per-beat detection noise. An arrhythmia supervisor classifies each beat by the ratio $\text{RR}_{\text{meas}}/\text{RR}_{\text{est}}$ and applies the matching reflex—*premature*: trust the measured phase, freeze the rate and learning; *missed*: trust the prediction, keep the rate—protecting both the loop and the template.

B. Algorithm summary

Per sample: (1) update trackers (10) and gate; (2) advance phase (11), correct at fiducials (12)–(13), run the supervisor; (3) ILC lookup/limited-subtract (7),(9) (update (8) on normal beats). The proposed online path is therefore constant-time per sample, with template memory linear in the window length.

V. COMPLEXITY ANALYSIS

Let L_w be the template length and C the channel count. The proposed phase-tracker + ILC path is lightweight: tracker updates and gating are constant-time, PI/PID phase update is constant-time (amortized), and ILC lookup/update is constant-time with memory L_w . Thus, per-sample complexity is $O(1)$ and memory is $O(L_w)$ for each channel. This profile is suitable for always-on low-power ear-EEG devices. For fairness, baseline adaptive cancellers (e.g., LMS/RLS) are reported separately in experiments and are not part of the proposed online deployment path.

VI. EXPERIMENTS

A. Real ear-EEG validation: coherence and repeatable-template evidence

To test the coherence assumption behind reference-driven cancellers and confirm that a genuine repeatable cardiac template exists outside simulation, we analyze a 10-minute segment (717 beats; 2 premature, 0 missed) from the SeizeIT2 dataset [12], which pairs a wearable behind-the-ear EEG channel (256 Hz) with a synchronized chest ECG lead during video-EEG epilepsy monitoring.

Coherence is near zero, even under best-case R-locked pooling. Fig. 2(a) shows magnitude-squared coherence between the ECG reference and the ear-EEG channel from long, many-average Welch segments across 0–60 Hz: coherence averages 6×10^{-10} over 8–20 Hz and is visually indistinguishable from zero across the whole spectrum – inconsistent with any stationary linear channel between $r(t)$ and $d(t)$. A skeptic could object that this washes out a relationship holding only locally around each beat; pooling short-time estimates over rolling 12-beat R-aligned batches (715 estimates, Fig. 2(b)) gives the same answer – coherence stays low and stable (0.06–0.17, mean 0.11) throughout the recording, an order of

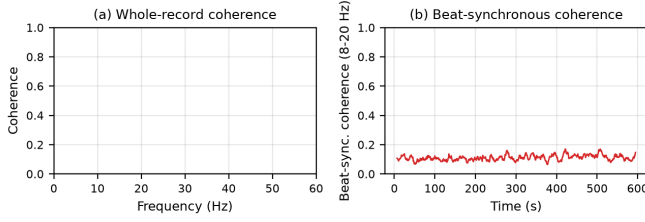


Fig. 2. Real ear-EEG/ECG coherence. (a) Whole-record Welch estimate (0–60 Hz): indistinguishable from zero across the spectrum. (b) Beat-synchronous estimate (8–20 Hz band), pooled over rolling 12-beat batches across a 10-minute recording (715 estimates): coherence stays low and stable even under best-case R-locked alignment.

magnitude below what a usable Hammerstein/RLS reference-driven canceller would need. Even granting the reference every possible timing advantage, the ear-EEG artifact waveform is not linearly predictable from the ECG waveform.

A repeatable, R-anchored template nonetheless exists. Coherence measures waveform predictability from the reference, not repeatability across beats. Non-causal ILC/synchronous averaging (§IV-B) applied directly to the real ear-EEG channel (Fig. 3) recovers a clear, consistent deflection time-locked to the ECG R-peak, confirming the cardiac field artifact at the ear is genuinely timing-repeatable – exactly the property ILC exploits – even though it fails the coherence test above. This is the real-data analogue of synchronous-average disk-servo RRO/NRRO separation: only the R-peak *instant* is used, never the ECG waveform shape. The causal-vs-buffered R-spike tradeoff is likewise real, not a simulation artifact: buffered matches non-causal (R-spike peak-to-peak 5.02×10^{-5} V vs. 5.07×10^{-5} V), while the causal template is capped by RR-prediction error to 3.47×10^{-5} V, a 32% reduction, consistent with §IV-B. Cardiac-band (1–20 Hz) power is reduced by 0.9/0.8/0.4 dB for non-causal/buffered/causal respectively; the modest magnitude versus the semi-synthetic ceiling (Table I) reflects broadband physiological noise at this electrode and that only the beat-locked component is addressable by ILC by construction.

These observations are the empirical basis for this paper’s scope: a null coherence result that rules out a reference-driven residual stage as a primary contribution here, and a positive repeatability result that justifies keeping the timing-driven ILC stage. The linear RLS canceller in Table I is retained only as a semi-synthetic baseline, evaluated on data explicitly given a known Hammerstein/volume-conduction channel between $r(t)$ and $d(t)$; it is not part of the proposed real-time deployment path.

B. Semi-synthetic data

The real-data evidence above has no ground truth $s(t)$, so we complement it with a semi-synthetic protocol that provides exact-reference SNR metrics; the ILC-driven design validated qualitatively above is here validated quantitatively. Following the semi-synthetic protocol common in EEG denoising [7], a low-CFA neural surrogate (pink noise plus an amplitude-

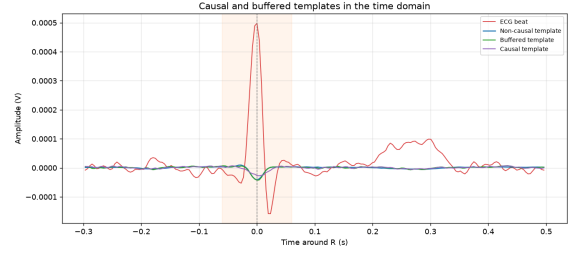


Fig. 3. Non-causal, buffered, and strict-causal templates learned from real ear-EEG (with one ECG beat shown only for R-peak timing reference). A clear, repeatable R-locked deflection is recovered despite the near-zero coherence of Fig. 2. Buffered matches non-causal; the causal (zero-latency) template’s R-spike is measurably damped by RR-prediction error, confirming the latency/suppression tradeoff of §IV-B on physiological data.

modulated 10 Hz alpha rhythm) provides ground truth $s(t)$. A synthetic ECG source $u(t)$ with RSA-modulated RR intervals is converted into a beat-locked artifact template with slow amplitude and morphology drift, then added at a controlled input SNR; one premature and one dropped beat are injected. This yields exact-reference metrics without manual labeling. Metrics: artifact SNR (steady state), spectral cardiac-peak reduction, and alpha preservation.

C. Baselines

Within the semi-synthetic protocol only (§VI-A shows why a reference-driven residual stage cannot contribute on the real recording), we compare against (i) a linear RLS/LMS ECG-reference canceller [3]–[5]; (ii) an EKF/state-space canceller [placeholder—to be added] [10]; and (iii) low-channel ICA [placeholder—to be added] [6], which is expected to degrade at $C < 8$.

D. Results

Table I reports steady-state artifact SNR for the main approaches, generated directly by the companion script to keep manuscript numbers synchronized with code. Fig. 4 shows single-pass online convergence for the three operating modes and the corresponding buffered-mode spectrum before and after cleaning. In single-pass online operation, buffered mode converges faster and to a higher final SNR than causal mode, consistent with the expected one-beat anchoring advantage. Spectrally, the cardiac comb is strongly reduced after cleaning while the 10 Hz alpha peak is preserved within about 10%, an implicit heartbeat-evoked-potential safeguard [11]. Ablations show that phase anchoring quality and mode selection (non-causal vs buffered vs strict-causal) dominate performance: buffered ILC is closest to non-causal performance, while strict-causal ILC is limited by RR prediction accuracy around sharp R-wave transitions.

Fig. 5 shows the time-domain cancellation over three consecutive beats in the converged regime. The synthetic reference $r(t)$ exhibits full PQRS morphology—P wave, QRS complex, and T wave—rather than a simplified sinusoidal surrogate, so the canceller must contend with a sharp, high-amplitude R deflection and a low-amplitude pre-R P wave. Af-

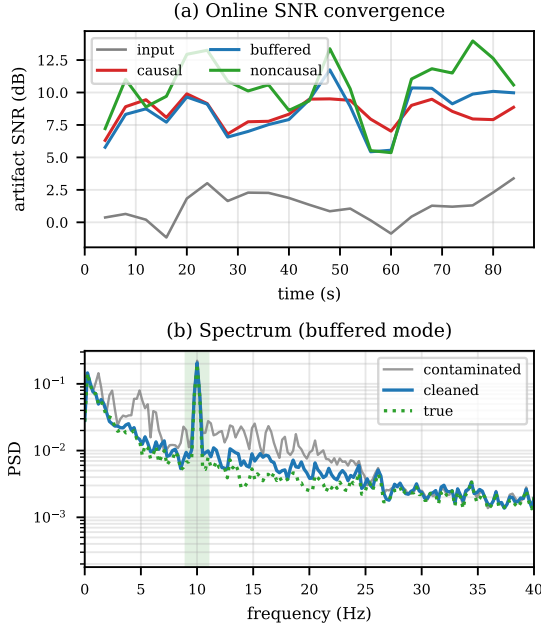


Fig. 4. (a) Online SNR convergence for the three modes (single causal pass). (b) Power spectrum before/after cleaning (buffered); shaded band marks the preserved 10 Hz alpha rhythm.

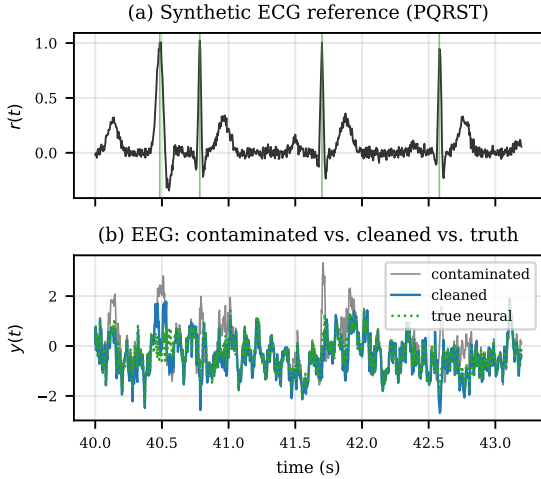


Fig. 5. Time-domain cancellation over three beats (converged, buffered mode). (a) Synthetic ECG reference $r(t)$ with full PQRST morphology; vertical lines mark detected R fiducials t_k . (b) Contaminated EEG $y(t)$, cleaned output, and ground-truth neural signal $s(t)$ over the same window.

ter processing, the cleaned EEG closely tracks the ground-truth neural signal $s(t)$: the R spike is suppressed by roughly an order of magnitude, the QRST tail is removed, and the residual P wave is attenuated, while the ongoing alpha oscillation is retained. This confirms that the phase-anchored template and residual canceller act on realistic cardiac morphology, not a reduced model.

TABLE I
STEADY-STATE ARTIFACT SNR (SEMI-SYNTHETIC; GENERATED).

Method	Latency	SNR (dB)
Input (contaminated)	–	1.27
Linear RLS ANC	causal	7.62
Proposed (causal)	0	8.58
Proposed (buffered)	1 beat	9.02
Proposed (non-causal)	offline	9.95

VII. CONCLUSION

We presented a control-theoretic synthesis—min/max-tracker PI-PLL phase estimation, ILC/repetitive template cancellation, and explicit non-causal/buffered/strict-causal operating modes—for real-time CFA removal in low-channel ear-EEG. On a real behind-the-ear EEG/chest-ECG recording, heartbeat-locked coherence between the two is near zero even under best-case R-locked pooling, while R-anchored synchronous averaging still recovers a clear, repeatable template and reproduces the causal-vs-buffered R-spike tradeoff quantitatively – evidence that, at least on this data, timing-only ILC is both necessary (reference-driven waveform regression has nothing to regress against) and sufficient (the artifact is genuinely timing-repeatable). On the semi-synthetic protocol, the method exceeds linear ANC at equal latency, exposes a quantified one-beat-latency vs. zero-latency trade-off set by RR predictability, preserves the alpha rhythm, and runs at constant per-sample cost suitable for edge deployment.

More broadly, this work aims to bridge mature mechatronic disturbance-rejection theory and biosignal denoising, opening a control-systems avenue for wearable neural interfaces. The disturbance-observer perspective extends naturally beyond the cardiac artifact: an inertial (gyroscope/accelerometer) reference could drive feed-forward cancellation of motion artifact, and an event-driven detector could trigger transient cancellation of eye-blink or jaw-clench artifacts, suggesting a unified, reference-driven, control-based artifact-rejection subsystem for earable EEG. Future work includes validation on recorded ear-EEG, multichannel spatial coupling for $C \leq 8$, an RSA-aware R predictor to lift the zero-latency R-spike ceiling, and an algorithm-unrolled low-power realization.

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